

Technical SEO, GEO, AEO & Performance Audit

heyppearl.io · Audit date: August 22, 2026 · Scope: full site, week-over-week

Executive Summary

This is the second full-site audit this quarter. The prior technical-SEO, PEARL-framework, and messaging/credibility audits have held — no regressions were found in canonical coverage, sitemap completeness, the P.E.A.R.L. definition, or the removal of fabricated case-study content. This week's pass surfaces a new, distinct layer of findings: AI-crawler configuration, internal-link integration of the two newest blog posts, FAQ/schema coverage on commercial pages, and a Core Web Vitals gap on hero images across the platform and solutions templates.

71/100 Overall Website Health	78/100 SEO	68/100 GEO (Generative Engine Optimization)
64/100 AEO (Answer Engine Optimization)	70/100 Technical SEO	72/100 Authority / EEAT
74/100 Content	62/100 Performance	80/100 Conversion

5 Critical issues, 6 High-priority improvements, 7 Medium-priority improvements, and 3 Low-priority items were found this week. None represent a full outage or de-indexing risk — the site is fundamentally healthy — but several are genuine, fixable inconsistencies between what the site publishes and what the site itself does.

Critical Issues

Problems that should be fixed immediately.

1. HeyPearl's own robots.txt has zero AI-crawler directives — direct contradiction of its own published advice

HIGH IMPACT

Finding: `src/app/robots.ts` allows all crawlers with a single blanket rule and names no AI bots at all (no GPTBot, ChatGPT-User, OAI-SearchBot, ClaudeBot, PerplexityBot, Google-Extended, Amazonbot, or Applebot-Extended). The site's newest blog post, "llms.txt and AI Crawler Access," instructs readers to write exactly these directives and calls their absence a foundational visibility gap. HeyPearl does not practice what that post preaches.

Why it matters: This is a self-referential credibility problem specific to an AI-authority company: any AI system, journalist, or prospective client who checks HeyPearl's own technical configuration against its own advice will find the advice unapplied. It also means HeyPearl is not making the deliberate training-vs-retrieval crawler distinction it recommends to clients.

Recommended fix: Add explicit Allow rules for OAI-SearchBot, PerplexityBot, ClaudeBot, and Google-Extended in `src/app/robots.ts`, following the exact pattern published in the site's own blog post.

2. No llms.txt file exists anywhere on the site

MEDIUM IMPACT

Finding: Confirmed absent from both the project root and `public/`. The same newest post explains what `llms.txt` is and how to write one; HeyPearl has never published its own.

Why it matters: Costs almost nothing to add, and turns a live inconsistency into a live proof point — HeyPearl's own `llms.txt` could be referenced directly from the blog post as a working example.

Recommended fix: Publish a genuine `public/llms.txt` describing HeyPearl's core pages, platform modules, and top authority content, following the exact structure the blog post recommends.

3. Both newest blog posts are structurally orphaned from the site's commercial and technical pages

HIGH IMPACT

Finding: Zero non-blog pages link to either `llms-txt-ai-crawler-access-guide` or `ai-visibility-local-service-businesses` (confirmed by direct grep). Neither is referenced from any `/platform/*` page (despite the crawler-access post being squarely technical-GEO content) nor from `/solutions/service-businesses` (despite the local-service post being a direct topical match for that page's audience).

Why it matters: Internal links are one of the strongest signals available for building topical authority around a new page. A post with zero inbound links from relevant commercial pages compounds far more slowly than one integrated into the site's link graph, and both AI engines and traditional crawlers weight internal link context heavily when assessing topical relevance.

Recommended fix: Add contextual links from `/platform/knowledge-graph`, `/platform/geo-engine`, or `/platform/authority-pages` into the `llms.txt` post, and from `/solutions/service-businesses` into the `local-service-business` post. Specific placements below in Internal Linking Recommendations.

4. PearlOS's two-part identity conflict was only reconciled on secondary pages — the canonical /platform/pearlos page still tells the narrower story

HIGH IMPACT

Finding: /platform/pearlos (the URL the nav, sitemap, and every internal link treat as PearlOS's authoritative page) defines PearlOS purely as a signal-orchestration OS, with zero mention of CRM, booking, or lead-nurture anywhere in the file. Meanwhile /about's FAQ content carries the fuller, reconciled definition (signals + CRM/booking automation). The most CRM-forward version of the definition now lives only in the dead, redirected /services/geo page and lib/services.ts.

Why it matters: A visitor who clicks through from any nav link, footer link, or the sitemap to the actual PearlOS product page gets a narrower definition than a visitor who happens to land on /about first. This is the exact kind of cross-page inconsistency that erodes both user trust and AI entity-consistency scoring, on the single highest-traffic PearlOS page on the site.

Recommended fix: Update /platform/pearlos's core definition paragraph to include the operational/CRM layer explicitly, matching the reconciled language already live on /about, so the canonical product page carries the complete definition.

5. /privacy's own canonical tag contradicts the sitemap that lists it as indexable

MEDIUM IMPACT

Finding: src/app/privacy/page.tsx explicitly sets its canonical to /policies — i.e., the page itself declares it is not the canonical version — yet sitemap.ts still lists /privacy as its own indexable URL at priority 0.3.

Why it matters: Sending search engines two conflicting signals about the same URL ("index this" via the sitemap, "don't — go here instead" via the canonical tag) is exactly the kind of contradiction crawlers are built to distrust, and it dilutes the very consolidation the canonical tag was meant to create.

Recommended fix: Remove /privacy from sitemap.ts's static pages array, since the page has already self-declared as non-canonical.

High-Priority Improvements

Changes with the highest expected impact after the critical fixes above.

1. Missing FAQPage schema on all 5 /solutions/* pages despite visible FAQ content already existing there

HIGH IMPACT

Finding: Each of /solutions/real-estate-agents, /brokerages, /teams, /service-businesses, and /enterprise renders a visible FAQ section, but none of the five emit FAQPage JSON-LD (confirmed via schema grep across src/app).

Why it matters: These are the highest commercial-intent pages on the site for each real-estate segment. FAQPage schema is one of the most reliable levers for featured snippets and AI-Overview-style answer boxes — leaving five ready-made FAQ sections unmarked is the single largest missed AEO opportunity found this week.

Recommended fix: Wrap each page's existing FAQ array in a FAQPage JSON-LD block, following the exact pattern already used on /faq and /about.

2. All 6 /platform/* module pages have no FAQ content at all — not a schema gap, a content gap

HIGH IMPACT

Finding: Direct grep confirms zero FAQ-related content on /platform/pearlos, /ai-visibility, /authority-score, /knowledge-graph, /geo-engine, and only a passing mention on /authority-pages. These are the pages most likely to be the target of a bottom-funnel "what is Authority Score / Knowledge Graph / GEO Engine" query.

Why it matters: Every one of these pages is a strong, specific candidate for 4-6 question-and-answer pairs answering exactly the definitional and comparison questions a prospect (or an AI engine composing an answer) would have about that module. This is content depth AND AEO opportunity in one gap.

Recommended fix: Add a 4-6 question FAQ block plus FAQPage schema to each of the 6 platform module pages, addressing what the module does, how it differs from the others, and how it fits into the P.E.A.R.L. sequence.

3. Hero images missing the priority attribute across roughly 10 high-traffic templates — a direct LCP risk

HIGH IMPACT

Finding: Confirmed zero priority attribute on the primary above-the-fold hero image on platform/pearlos, platform/ai-visibility, and solutions/real-estate-agents (and, per the same shared component pattern, likely all remaining /platform/* and /solutions/* pages). Only the homepage, /insights/[slug], and /about/founder currently set priority.

Why it matters: Next.js Image defaults to lazy-loading. An above-the-fold hero image without priority is not preloaded, which directly and measurably delays Largest Contentful Paint — the single most heavily weighted Core Web Vital in Google's ranking systems — on exactly the pages most likely to be a paid-traffic or AI-citation landing destination.

Recommended fix: Add the priority prop to the primary hero Image component on every /platform/* and /solutions/* page.

4. images.unoptimized: true disables Next.js's built-in image optimization site-wide

HIGH IMPACT

Finding: next.config.js sets images.unoptimized: true, which turns off automatic resizing, WebP/AVIF format conversion, and responsive srcset generation for every next/image component on the site — even though 100% of images already correctly use next/image (no raw HTML img elements found).

Why it matters: This is likely the single largest aggregate performance lever on the site. The lazy-loading and layout-shift protection of next/image is preserved, but every image is served at its original file size and format regardless of viewport — a real, sitewide Core Web Vitals cost that compounds across dozens of pages.

Recommended fix: Re-enable Next.js image optimization (remove or scope down the unoptimized flag) if the current hosting/deployment target supports it, or move to a CDN-based image optimization layer if the flag was set for a specific infrastructure reason.

5. BreadcrumbList schema is missing from roughly 11 template types

MEDIUM IMPACT

Finding: Confirmed present only on /insights/[slug] and /faq/[slug]. Absent from /resources/[slug], all 6 /platform/* pages, and all 5 /solutions/* pages.

Why it matters: Breadcrumb rich results improve SERP click-through and give AI engines an explicit, structured signal of where a page sits in the site's topical hierarchy — directly relevant to GEO entity/topic-clustering signals, not just a cosmetic SERP feature.

Recommended fix: Add a BreadcrumbList JSON-LD block to /resources/[slug], and to each /platform/* and /solutions/* page, following the pattern already implemented on /insights/[slug].

6. /services/* pages are dead code carrying stale, conflicting content that no visitor or crawler should ever see, but which still exists in the repository

MEDIUM IMPACT

Finding: /services, /services/geo, /services/heylocal, and /services/authority-websites are all permanently 301-redirected away in next.config.js, yet the underlying page files still exist, still render real content, and — in services/geo's case — still carry the older, more CRM-forward PearlOS definition that was never updated when the reconciliation work happened on live pages.

Why it matters: Dead pages with stale definitions are a maintenance trap: the next person to search the codebase for "PearlOS definition" will find four different versions across live and dead files with no way to tell which is current. It also means real work (writing/updating that copy) is sitting in files no user will ever read.

Recommended fix: Delete the four /services/* page files entirely now that their redirects are permanent and stable, or explicitly repurpose them if there's a future plan to un-redirect them.

Medium-Priority Improvements

1. /policies and /privacy render identical title tags

MEDIUM IMPACT

Finding: Both pages set the exact title "Privacy Policy," producing duplicate title tags across two separate, both still-crawlable URLs.

Recommended fix: Differentiate the /privacy title (e.g., a note that it redirects/points to the canonical policy) or accept the duplication is low-risk given the canonical tag already resolves it — but a quick differentiation removes any ambiguity.

2. /terms-of-service has no canonical tag

MEDIUM IMPACT

Finding: Unlike /privacy and /policies, /terms-of-service's metadata sets only title and description — no alternates.canonical.

Recommended fix: Add alternates: { canonical: 'https://heyppearl.io/terms-of-service' } to match the pattern used on the site's other legal pages.

3. /terms and /privacy-policy use temporary (307) redirects instead of permanent (301)

MEDIUM IMPACT

Finding: Both are implemented as Next.js redirect() calls inside the page component, which issue a 307 Temporary Redirect — unlike the /services/* redirects, which correctly use next.config.js's permanent redirect map (301).

Recommended fix: Move both redirects into next.config.js's redirects() array with permanent: true, consistent with the rest of the site's redirect strategy, so link equity consolidates correctly.

4. A hardcoded www subdomain link breaks domain-identity consistency

LOW IMPACT

Finding: /terms-of-service links to https://www.heyppearl.io/privacy-policy, while every canonical tag, Open Graph URL, and metaDataBase reference elsewhere on the site uses the bare apex domain heyppearl.io.

Recommended fix: Change the link to a relative path (/privacy-policy) or the apex domain, matching the rest of the site.

5. Primary navigation's "Solutions" dropdown does not surface any of the 5 actual /solutions/* pages

MEDIUM IMPACT

Finding: The Solutions dropdown's children link to GEO Farming AI (external), PearlOS, AI Authority, AI Operations, and AI Visibility — all /platform/* URLs. None of the five pages actually named at /solutions/real-estate-agents, /brokerages, /service-businesses, /teams, or /enterprise appear in the dropdown, even though the parent "Solutions" label does correctly link to /solutions itself, and the footer links all five directly.

Recommended fix: Add the five /solutions/* pages to the Solutions dropdown menu (or restructure the dropdown to segment by audience rather than by platform module), so the nav's most-used surface matches what "Solutions" actually means on this site.

6. /pricing is dead code carrying a now-meaningless canonical tag

LOW IMPACT

Finding: The page file exists in full, with its own canonical tag and complete pricing-tier content, but next.config.js redirects /pricing to /get-started before the page can ever render for a real visitor or crawler.

Recommended fix: Either delete the page file since it's unreachable, or remove the redirect if publishing pricing is the intended direction — leaving both in place serves no one.

7. Founder bio sidebar blurb is thin relative to the credentials referenced elsewhere on the site

LOW IMPACT

Finding: The blog sidebar author card's bio line is a short, generic sentence, while /about/founder describes a fuller professional history (brokerage/brand-scaling experience). The two are not required to match, but the shorter version is a missed EEAT opportunity on every single blog post.

Recommended fix: Expand the sidebar bio to include one concrete credential or specific experience marker, rather than only a generic mission statement.

8. H2 headings on /platform/* and /solutions/* pages are written as benefit statements, not questions

LOW IMPACT

Finding: Unlike the blog content (which frequently uses direct question-format H2s, e.g. "Does blocking AI crawlers protect your content, or hurt your visibility?"), the platform and solutions templates use declarative headings throughout.

Recommended fix: Where natural, convert 1-2 H2/H3 headings per platform/solutions page into direct question format, to improve featured-snippet and voice-search matchability on commercial pages, not just blog content.

Low-Priority Improvements

1. /privacy-policy is linked from 2 places (ContactForm.tsx, terms-of-service page) instead of linking directly to the canonical /policies

LOW IMPACT

Fix: Update both links to point straight to /policies, removing an unnecessary redirect hop.

2. CSP allows 'unsafe-inline' for scripts

LOW IMPACT

Required today for the inline GA4/Meta Pixel bootstrap scripts, but worth revisiting with nonce-based CSP if those scripts are ever refactored.

Fix: No immediate action required; flagged for awareness, not a ranking or visibility factor.

3. Internal working documents sit at the repository root

LOW IMPACT

AUTHORITY_PAGE_PROMPT.md, DEPLOY_PLAYBOOK.md, IMAGE_PROMPTS.md, and blog-articles.json are not served publicly and carry no SEO risk, but are housekeeping clutter.

Fix: Move to a /docs or /internal folder if repo hygiene matters to the team; zero user-facing impact either way.

New Content Recommendations

Recommendation	Why	Impact
Platform module FAQs (6 pages)	Add a 4-6 question FAQ block to each /platform/* page — see High-Priority Improvements.	High
A dedicated "PearLOS vs. a traditional CRM" comparison page or section	Directly addresses the PearLOS identity conflict by explaining, in the customer's own words, how PearLOS's signal-orchestration layer relates to the CRM/booking automation layer it also runs — turns a current inconsistency into a clear differentiator.	High
A public llms.txt file	See Critical Issues — doubles as both technical infrastructure and a citable proof point from the site's own blog content.	Medium
Brokerage-segment case study	The site's real case-study library (src/data/results.ts) currently has zero entries in the "Brokerage" segment, only Independent Agent and one Team story — the /solutions/brokerages page has no real, segment-matched proof to point to.	Medium
Short comparison article: "GPTBot vs. OAI-SearchBot vs. PerplexityBot — Which AI Crawlers Should You Actually Block?"	A natural, citable follow-up to the llms.txt post that AI engines and technical readers alike would find genuinely useful, and reinforces the technical-authority cluster.	Medium

Internal Linking Recommendations

From	To	Notes
/platform/knowledge-graph or /platform/geo-engine	→ /insights/llms-txt-ai-crawler-access-guide	Add a contextual link (e.g., "Before AI engines can index this signal, they need to be able to crawl your site — here's how to configure that") — closes the orphaned-post gap on the most topically relevant platform page.
/solutions/service-businesses	→ /insights/ai-visibility-local-service-businesses	Add a contextual link from the page's FAQ or body copy into the new local-service-business playbook — direct audience match, currently zero inbound links exist.
/platform/pearlos	→ /about (or vice versa)	Cross-link the two pages carrying PearLOS's definition so a visitor (and a crawler) can see both halves of the system are the same product, reducing the entity-consistency risk until the definitions are fully merged.
Primary nav "Solutions" dropdown	→ all 5 /solutions/* pages	See Medium-Priority Improvements — this is a navigation fix as much as an internal-linking one, but it is the highest-value internal link opportunity on the site by traffic exposure.

/results (or /results/case-studies)	→ each /solutions/* page's proof section	Once a brokerage-segment case study exists, link it specifically from /solutions/brokerages rather than relying on the generic proof language currently there.
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Schema Recommendations

Schema Type	Where	Priority
FAQPage	All 5 /solutions/* pages	High — visible FAQ content already exists, schema is the only missing piece
FAQPage + new content	All 6 /platform/* pages	High — requires writing the FAQ content first, then wrapping it
BreadcrumbList	/resources/[slug], all /platform/*, all /solutions/*	Medium — straightforward to add, follows the existing /insights/[slug] pattern exactly
Service schema	Consider for /platform/* module pages	Low/Medium — each module (Authority Pages, Authority Score, Knowledge Graph, AI Visibility, GEO Engine) is a distinct, describable service offering and is a natural Service schema candidate not currently used anywhere on the site

GEO Recommendations

- Add explicit Allow directives for OAI-SearchBot, PerplexityBot, ClaudeBot, and Google-Extended in robots.ts — turn the site's own published advice into the site's own practiced configuration.
- Publish a real public/lms.txt describing HeyPearl's core pages and top authority content, and reference it as a working example from the lms.txt blog post itself.
- Reconcile the PearlOS definition on its canonical /platform/pearlos page so AI systems encountering that page get the complete, consistent entity description rather than the narrower of two versions.
- Add a brokerage-segment case study so /solutions/brokerages has entity-consistent, segment-specific proof content an AI engine could cite for brokerage-specific queries — currently the segment has no matching real case study.
- Cross-link the two newest blog posts into their most topically relevant commercial pages (see Internal Linking Recommendations) — internal link context is a meaningful topical-authority signal for generative engines assessing what a page is really about.

AEO Recommendations

- Wrap the existing FAQ content on all 5 /solutions/* pages in FAQPage JSON-LD — highest-ROI single AEO action available this week.
- Write and publish a 4-6 question FAQ block for each of the 6 /platform/* pages, addressing what each module is, how it differs from the others, and how it fits the P.E.A.R.L. sequence — direct answer-engine and featured-snippet targets.
- Convert 1-2 declarative H2 headings per /platform/* and /solutions/* page into direct question format, to improve voice-search and featured-snippet matchability on commercial pages, not only blog content.

- Add BreadcrumbList schema to the 11 template types currently missing it, giving AI engines an explicit structural signal of topic hierarchy.

Technical SEO Recommendations

- Add priority to the primary hero Image component on every /platform/* and /solutions/* page — direct, measurable LCP improvement.
- Re-evaluate images.unoptimized: true in next.config.js — this is likely the single largest aggregate Core Web Vitals lever available on the site.
- Remove /privacy from sitemap.ts since the page's own canonical tag already declares it non-canonical.
- Add a canonical tag to /terms-of-service.
- Move the /terms and /privacy-policy redirects from in-page redirect() calls into next.config.js's permanent redirect map, consistent with the rest of the site.
- Differentiate the duplicate "Privacy Policy" title tags on /policies and /privacy.
- Delete the four dead /services/* page files now that their redirects are permanent and stable.

EEAT Recommendations

- Reconcile PearlOS's definition on its own canonical product page (/platform/pearlos) — entity consistency is itself an EEAT and GEO signal, and this is the most visible remaining inconsistency on the site.
- Expand the blog sidebar author bio to include one concrete credential or specific experience marker, rather than only a generic mission statement — every post on the site currently carries the same thin version.
- Add a brokerage-segment case study to close the one clear gap in the real case-study library's segment coverage.
- Consider a short "Methodology" or "How We Measure Results" page linked from /results, giving both human readers and AI systems a single authoritative source for how HeyPearl's metrics and case studies are verified — strengthens Trustworthiness specifically.

Performance Recommendations

- Add priority to above-the-fold hero images on all /platform/* and /solutions/* pages (High impact, low effort).
- Re-evaluate whether images.unoptimized: true is still necessary; if the hosting target supports Next.js image optimization or a CDN-based equivalent, re-enabling it is the highest-leverage sitewide performance change available (High impact, moderate effort).
- No render-blocking scripts were found — GA4, Meta Pixel, and third-party scripts all correctly use next/script with afterInteractive; no action needed there.
- Font loading via next/font/google with explicit fallback stacks is already correctly configured; no action needed.

Conversion Recommendations

- Add the brokerage-segment case study to /solutions/brokerages so that page's CTA is backed by segment-matched proof rather than only generic language — proof proximity to CTA is a conversion lever, not just a content one.
- Cross-link the new local-service-business blog post from /solutions/service-businesses; a reader who lands on that page mid-research and finds a directly relevant, deeper resource is more likely to convert than one who hits a dead end.
- No CTA, form, or navigation defects were found that would need softening or removal for SEO reasons — the existing calls-to-action are consistent and clearly matched to each page's offer.

Weekly Action Checklist

Sorted by highest ROI first — priority, then estimated effort, then expected impact.

Task	Priority	Est. Time	Impact
☐ Add explicit AI-crawler Allow directives to robots.ts	Critical	15 min	High
☐ Remove /privacy from sitemap.ts	Critical	5 min	Medium
☐ Add 1-2 internal links from /platform/* and /solutions/service-businesses into the 2 newest blog posts	Critical	15 min	High
☐ Reconcile PearlOS definition on /platform/pearlos to match the fuller version already live on /about	Critical	20-30 min	High
☐ Publish a real public/lms.txt	Critical	30 min	Medium
☐ Add priority to hero images on all /platform/* and /solutions/* pages	High	30-45 min	High
☐ Wrap existing FAQ content in FAQPage schema on all 5 /solutions/* pages	High	30-45 min	High
☐ Delete the 4 dead /services/* page files	High	10 min	Medium
☐ Write and schema-tag FAQ content for the 6 /platform/* module pages	High	2-3 hrs	High
☐ Add BreadcrumbList schema to /resources/[slug], /platform/*, /solutions/*	High	1-2 hrs	Medium
☐ Re-evaluate images.unoptimized in next.config.js	High	Investigation + testing	High
☐ Add canonical tag to /terms-of-service	Medium	5 min	Medium

[] Move /terms and /privacy-policy redirects into next.config.js as permanent 301s	Medium	10 min	Medium
[] Differentiate duplicate title tags on /policies and /privacy	Medium	5 min	Low
[] Add the 5 /solutions/* pages to the primary nav Solutions dropdown	Medium	20 min	Medium

This audit reflects the codebase as of the date above. Next week's pass should re-verify every Critical and High item here was actually resolved (not just attempted), and re-run the AI-crawler-access and internal-linking checks specifically, since those are the categories most likely to silently regress as new content is added.